

Course code: ME343

Course Name: Non-Conventional Energy Resources

Pre-requisite: Thermodynamics, Fluid Mechanics, Heat Transfer

Objective of the course:

To provide elementary knowledge about various non conventional energy sources

Course Content:

Unit 1 (7 lectures)

Solar Radiation: Direct and diffused radiation, Radiation measuring instrument

Unit 2(8 lectures)

Solar Collectors: Optics of Solar collectors, Flat plate collectors, Collector Design and performance, Selective surface, Collector materials.

Unit 3 (6 lectures)

Applications: Water heating, Steam generation, Space heating and cooling, Distillation, Drying, Solar ponds.

Unit 4 (5 lectures)

Photo-Voltaic Conversion: Thermo-electric and thermo ionic solar cells, Performance

Unit 5(5 lectures)

Bio Energy: Photosynthesis Conversion of plant materials and waste to fuels.

Unit 6(5 lectures)

Wind Energy: Windmills, Blade design, Application.

Unit 7(3 Lectures)

Fuel cells : Types of fuel cells, performance and applications

Laboratory Component: No

Readings:

Text Books:

1. S.P. Sukhatme, Solar Energy, Tata Mcgrawhill
2. S. Ahmad, Wind Energy, PHI
3. H.S. Mukund, Clean Energy and Fuels from Biomass, Wiely
4. B.H. Khan, Renewable Energy, TMH